

2021

INSIDER LANDSCAPE REPORT



INTRODUCTION

The 2021 Insider Landscape Report reveals the latest trends and challenges facing organizations in this new environment. The report explores how IT and cybersecurity professionals deal with risky insiders and how organizations are preparing to better protect their critical data and IT infrastructure.

Key findings include:

- Security professionals are more concerned about inadvertent (71%) or negligent (60%) data leaks than malicious data breaches (52%)
- 52% of people find it more difficult to detect and prevent insider attacks than external cyber attacks
- 62% of people are concerned or very concerned about insider risk
- Only 29% of organizations currently use an insider risk management solution, despite being more concerned about accidental data leaks
- 65% of budgets will either stay the same or decline

This 2021 Insider Landscape Report has been produced by Cybersecurity Insiders, the 500,000-member community for information security professionals to explore how organizations respond to the evolving security threats in the cloud. We would like to thank [Code42](#) for supporting this unique research.

We hope you find this report informative and helpful as you continue your efforts in protecting your IT environments against insider threats.

Thank you,

Holger Schulze



Holger Schulze

CEO and Founder
Cybersecurity Insiders

Cybersecurity
INSIDERS

TYPES OF INSIDER THREATS

At its core, insider risk is a data protection problem. Security professionals agree, 71% see accidental or negligent data leaks as the biggest insider threats.

► What type of insider threats are you most concerned about?



71%

**Inadvertent
data breach/leak**

(e.g., careless user causing
accidental breach)



60%

**Negligent
data breach**

(e.g., user willfully ignoring
policy, but not malicious)



52%

**Malicious
data breach**

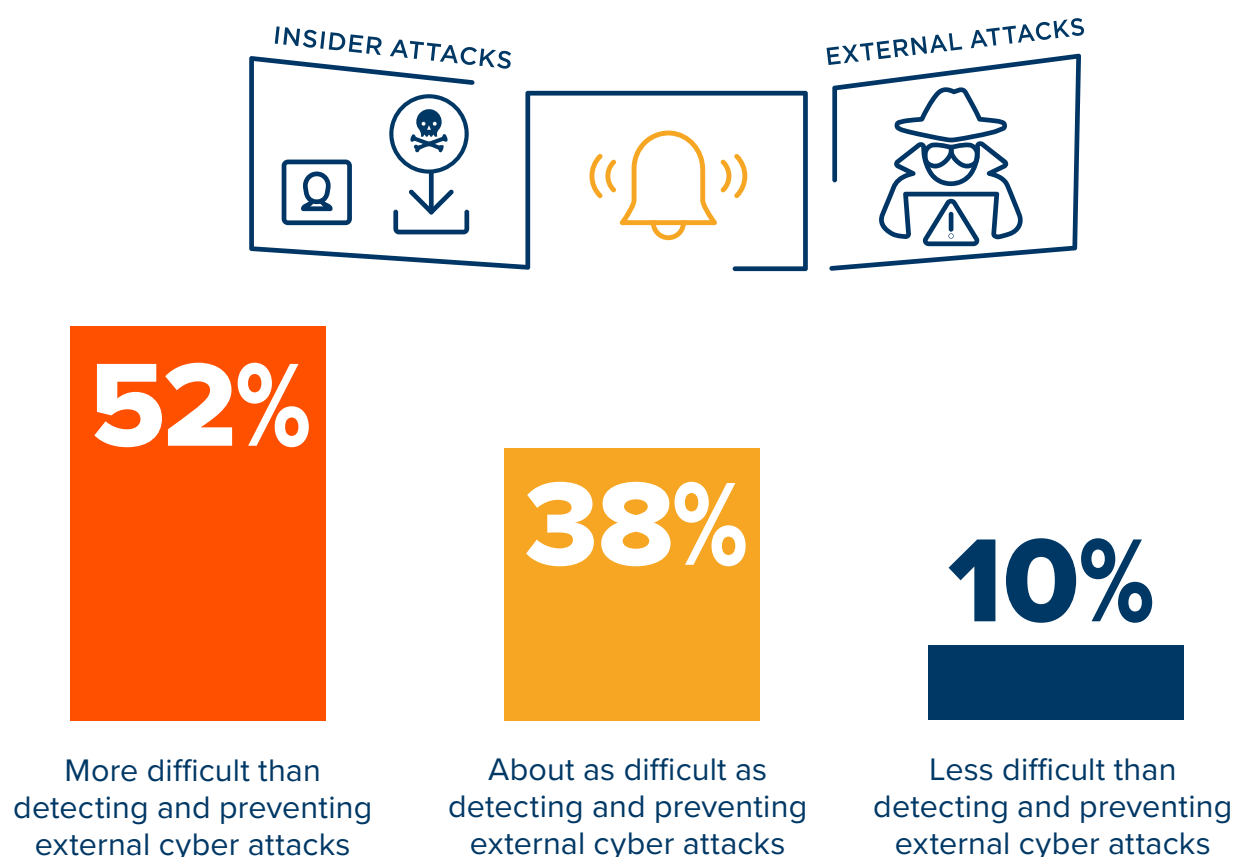
(e.g., user willfully
causing harm)

INTERNAL VS. EXTERNAL ATTACKS

Data security breaches have become much more difficult to assess. Trying to block all risky activities can have a huge impact on productivity and innovation. In order to successfully address insider risk, organizations must shift from policy-based to risk-based cybersecurity approaches. To make this transition, companies need to consider their insider risk tolerance — the delicate balance between protecting valuable data and enabling speed, collaboration and innovation.

When comparing internal attacks to external cybersecurity attacks, 52% of respondents confirm that internal attacks are more difficult to detect and prevent than external cyber attacks. Since insiders have approved access privileges, it can be challenging to distinguish legitimate use cases from malicious attacks.

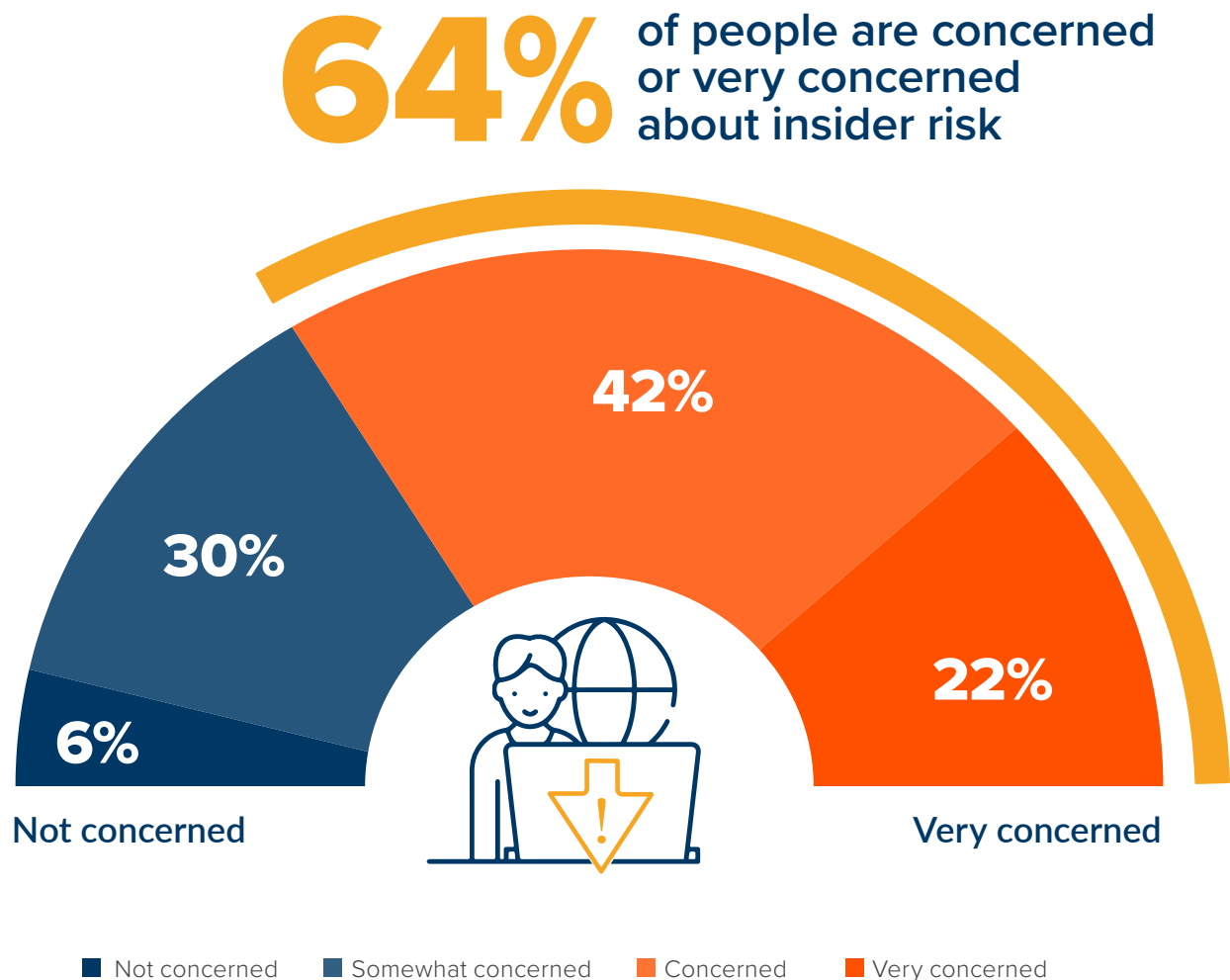
► How difficult is it to detect and prevent insider attacks compared to external cyber attacks?



INSIDER RISK CONCERNS

The rapid shift to remote work due to the pandemic increased the problem of insider risk - and with the hybrid workforce likely here to stay post-pandemic, 64% of people are concerned or very concerned about insider risk.

- ▶ **With a post-COVID return to the office imminent, or a new trend of a more hybrid workforce being likely, how concerned are you about insider risk?**



INSIDER THREAT IMPACT

Operational disruptions or outages (41%), continues to be one of the top three challenges, followed by loss of critical data (39%) and brand damage (31%).

► What impact have insider threats had on your organization?



41%

Operational
disruptions
or outages



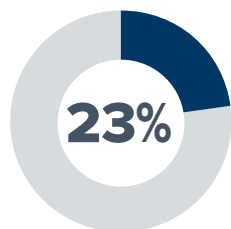
39%

Loss of
critical data

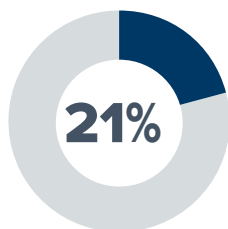


31%

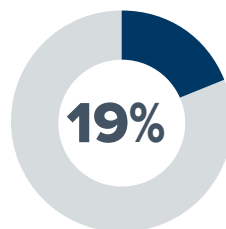
Brand damage



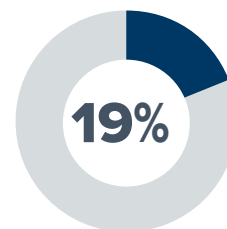
No impact



Legal liabilities



Expenditure
remediating successful
intrusions



Loss in
competitive edge

Loss in revenue 19% | Non-compliance with regulations 14% | Loss in market valuation 14%

MONITORING USER DATA

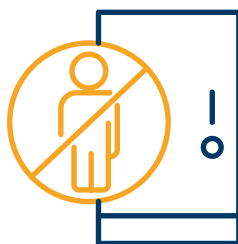
Monitoring insider risk indicators is essential to ensure an effective security posture. Sixty-two percent of security professionals claim it is important to monitor content data of employees with access to sensitive data, 57% perceive it is important to track data during a terminating notice, and 31% believe it is important to focus on employee type, like contractors, new hires, and interns.

► What context data about users is important to monitor for insider threats?



62%

**Employees with
access to sensitive data**
(IP, source code, trade secrets)



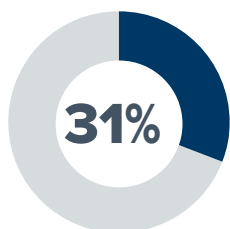
57%

**Employment
termination**
(Notice period)



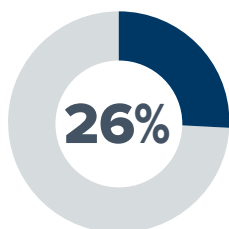
31%

Employee type
(contractor, new hire,
intern)



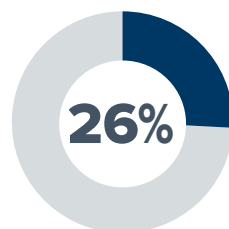
31%

Employees with
unsatisfactory
performance reviews



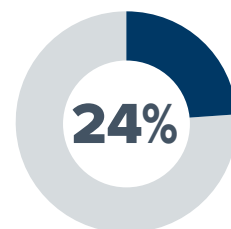
26%

Time of day IT
resources are
accessed



26%

Prior infractions
of risky behavior



24%

Unusual uploads to
off-site storage

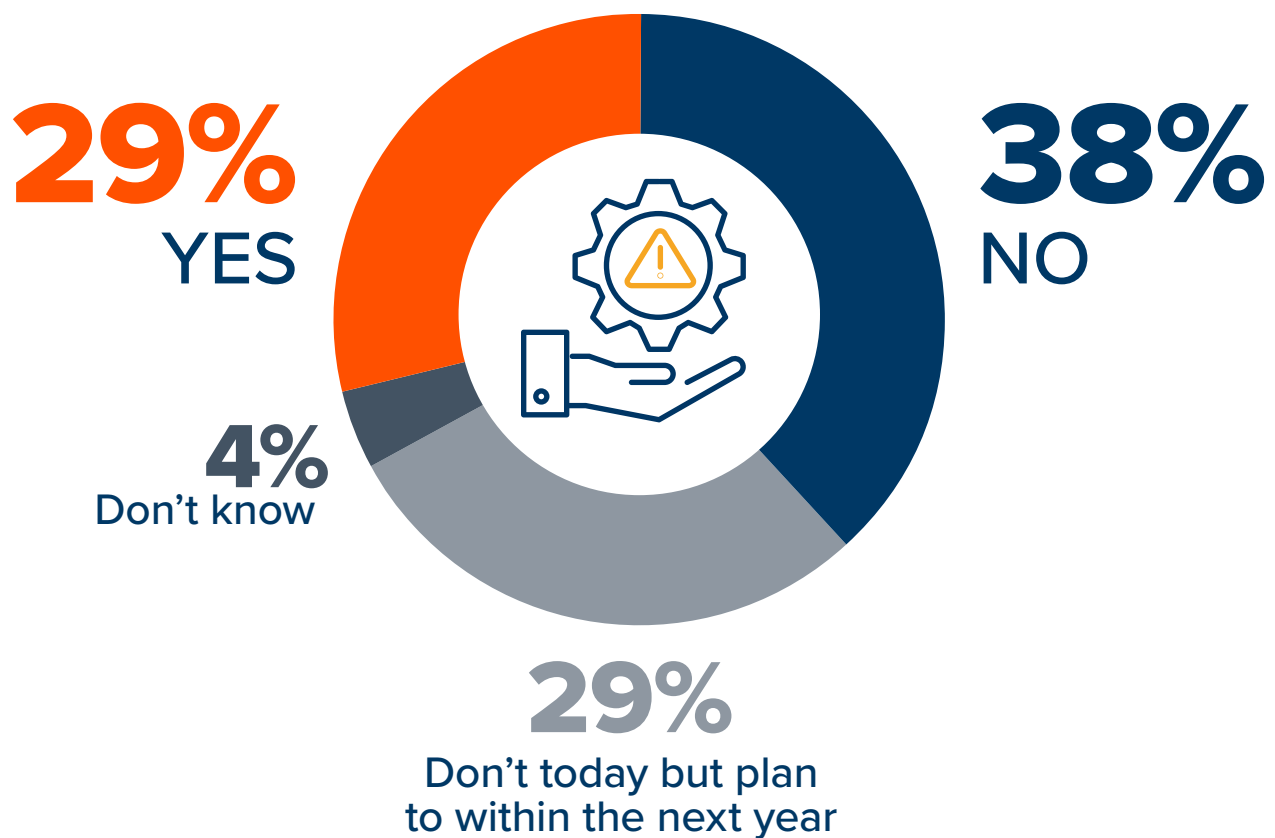
Job seekers (actively using job search sites) 24% | Spike in activity 24% | Privilege escalation 23% | Geo-location anomaly 22% | Badging data 21% | Email to unusual external domains 20% | Vacation schedule 19% | Access from previously unseen/unusual device 18% | Website visits (job sites, etc.) 17% | Expense reports 15% | Hotline calls 12% | Peer group behavior 10% | Sentiment analysis 6% | Other 4%

RISK MANAGEMENT SOLUTION

As the world of work keeps accelerating, the same empowering ways of working that will define tomorrow's success also drive today's rapidly growing insider risk to data. We can't ignore the need for a new data security paradigm to fit the future world of work.

Insider Risk Management offers a data-centric approach, which ensures compliance with data use policies, builds a more risk-aware culture, and speeds the time to security value. However, it is interesting to note, only 29% of organizations currently use an Insider Risk Management solution, despite being more concerned about accidental data leaks.

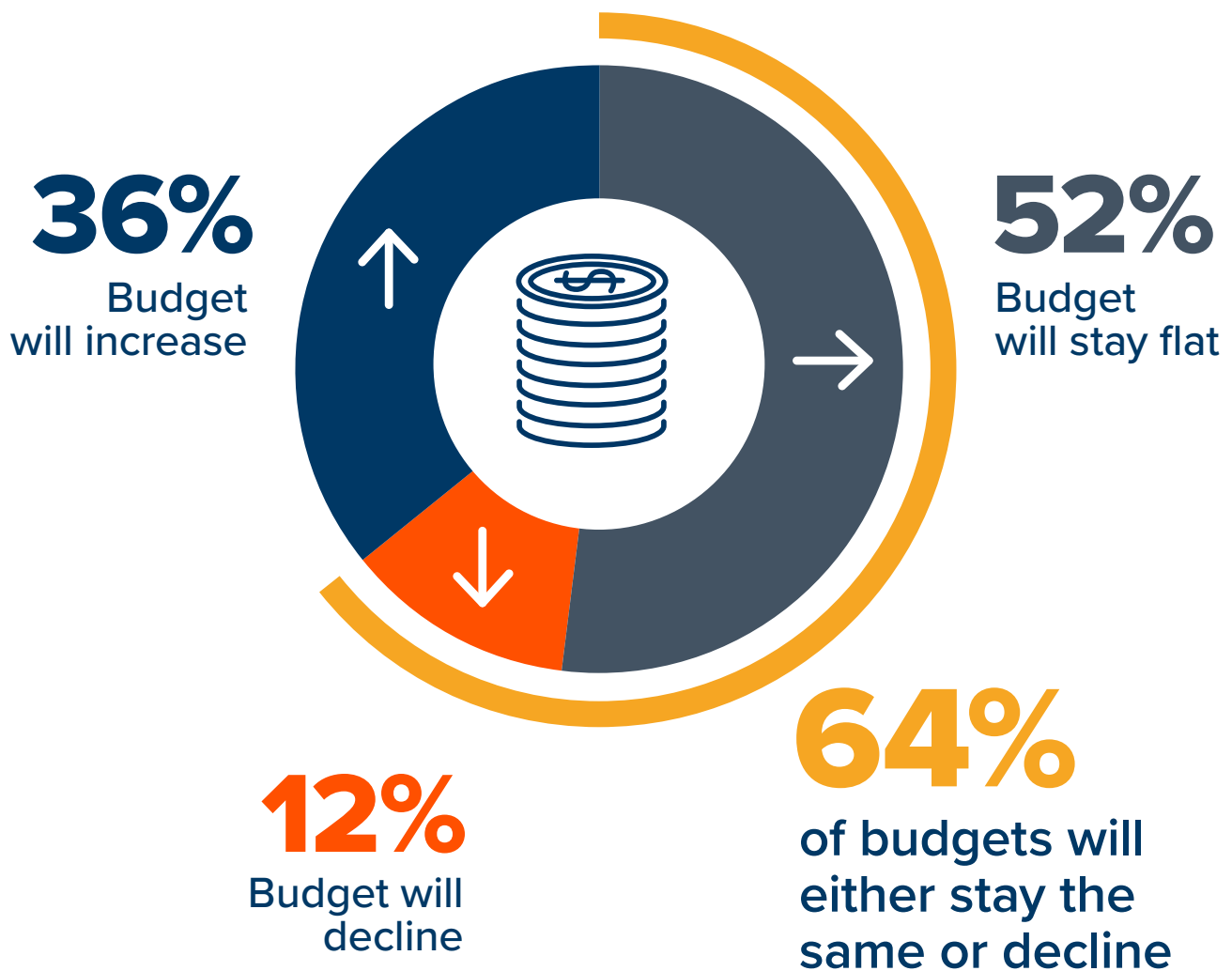
► Does your organization use an Insider Risk Management solution?



CLOUD SECURITY BUDGET

During the pandemic, there have been countless watershed moments in the business world. Savvy companies recognize that enabling a fast-paced, cloud-powered collaboration culture is essential to thriving companies in the new business environment. Achieving this goal hinges on a company's ability to manage the exponentially growing insider risk that these new ways of working introduce. Despite this understanding, 64% of security professionals claim that their budgets will either remain flat or decline.

► How is your security budget changing over the next 12 months?



INSIDER THREAT PROGRAM DRIVERS

The creation of formal insider threat programs is typically driven by information security governance programs (42%), followed by regulatory compliance and proactive security team initiatives (tying at 41%), rather than a response to insider incidents.

Interestingly, only 12% of security leaders say that a directive from the board of directors was a driver behind the creation of an insider threat program, indicating a lack of recognition of the connection between managing insider threats and organizational health.

► What is the primary driver of your insider threat program?



42%

Information security governance programs



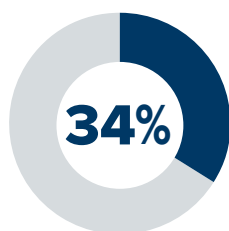
41%

Regulatory compliance

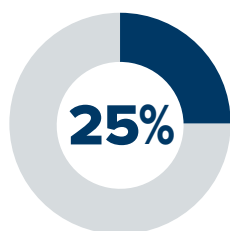


41%

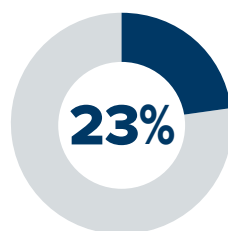
Proactive security team initiatives



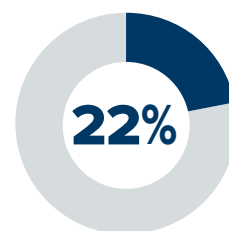
Proactive IT team initiative



Incident(s) that impacted peers or relevant industry



Directive from the executive management team



Suspected incident

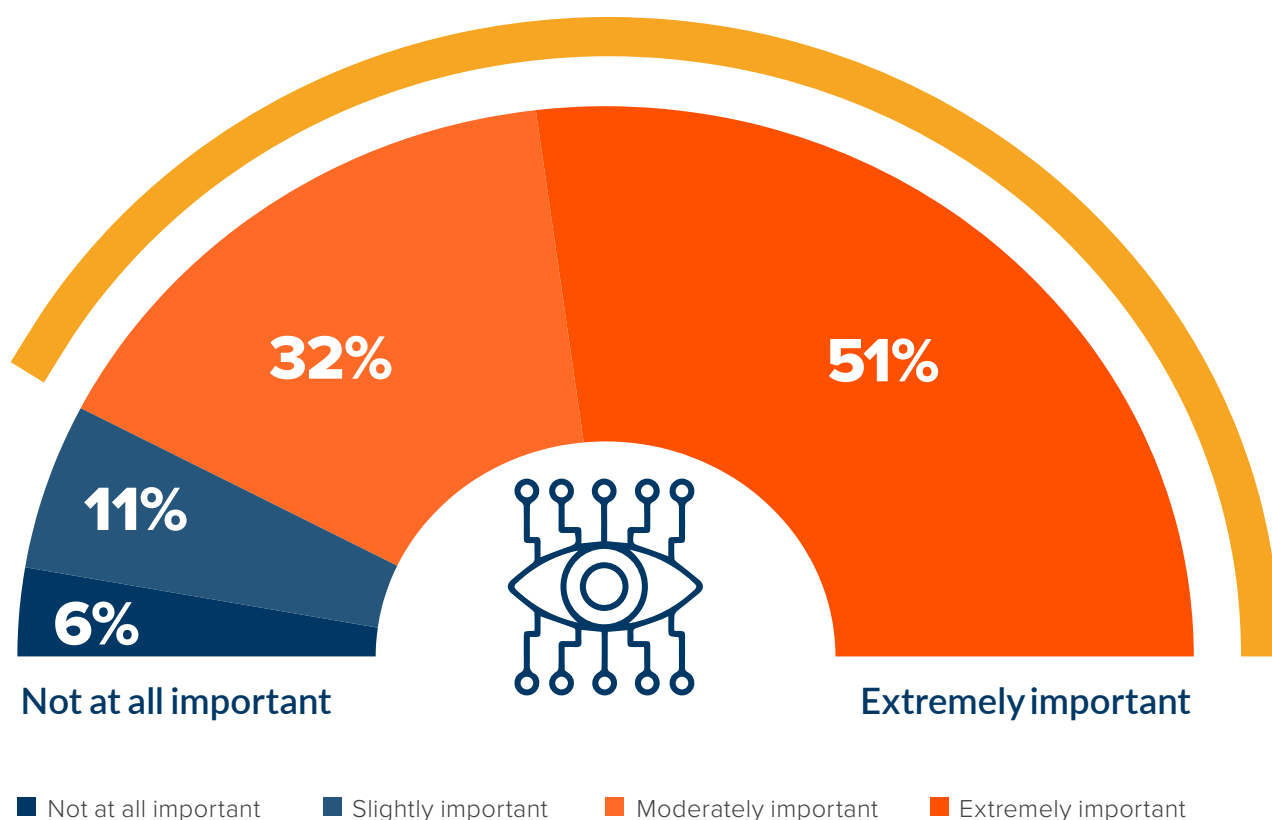
Previously confirmed incident 21% | Directive from the board of directors 12% | We do not have an insider threat program 12% | Other 4%

IMPORTANCE OF UNIFIED VISIBILITY

Visibility and control are paramount in preventing an insider threat. Almost all organizations (83%) consider unified visibility and control across the entire environment – on premises and in the cloud – moderately to extremely important.

- ▶ When it comes to insider threats, how important is unified visibility and control across your entire environment – on premises and in the cloud?

83% consider unified visibility and control across the entire environment – on premises and in the cloud – moderately to extremely important



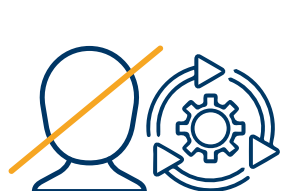
TOOLS TO COMBAT INSIDER THREATS

Lack of suitable tools to combat insider threats and protect sensitive data does not seem to be a perceived issue. A majority (63%) of organizations confirm they feel the right tools exist for keeping track of sensitive data and protecting it against insider threats. However, conventional, policy-based approaches like DLP, CASB, and UEBA focus on compliance, offering a perception of protection at best. When blocking is relied upon as the defacto response, the organization suffers in terms of employee productivity and the security team's internal reputation. With 42% of organizations thinking they lack the staff to implement existing tools, clearly productivity at many organizations is being perceived as a challenge.

- Do you feel the right tools exist for keeping track of your data and keeping your sensitive data, like intellectual property, safe from insider threats?



- If yes, what prevents you from implementing the existing tools?



42%

Lack the staff to implement



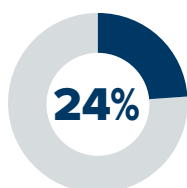
37%

Tools like DLP are too expensive

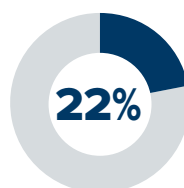


32%

Don't have the staff to maintain DLP tools



Data classification is too cumbersome



Deployment challenges related to remote or personal devices

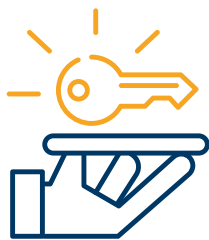
Tools inhibit productivity 14% | Other 19%

RISKY INSIDERS

Given the diversity of users that pose an insider threat, security teams need the ability to prioritize the risks that matter most, while also being able to see everything - all data activity, across all vectors and all users – on and off the network. To accomplish this, they need to see deeper context so they can parse risky activity and users from legitimate work. They need the ability to orchestrate highly automated, right-sized responses. Most of all, they need to commit to protecting the business, without sacrificing speed, collaboration, and innovation among users.

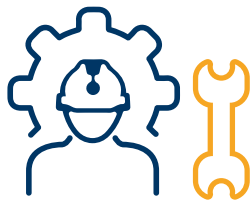
The survey indicates that privileged IT users/admins (55%), contractors (53%), privileged business users (51%), and regular employees (49%) pose the biggest insider security risk to organizations.

► What type(s) of insiders pose the biggest security risk to organizations?



55%

Privileged IT
users/admins



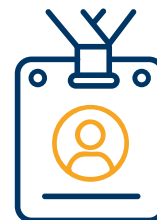
53%

Contractors/
service providers/
temporary
workers



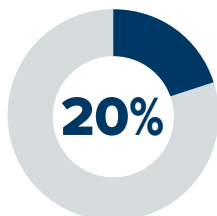
51%

Privileged
business
users/
executives

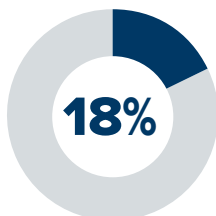


49%

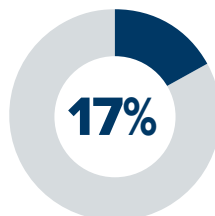
Regular
employees



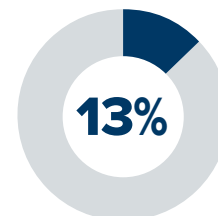
Executive
managers



Other IT
staff



Business
partners



Customers/
clients

Interns 3% | Not sure/other 5%

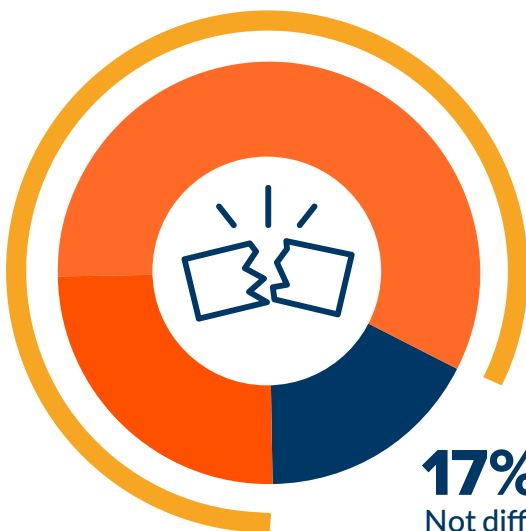
COSTLY INSIDER ATTACKS

Eighty-three percent of companies find it moderately to very difficult to determine the damage incurred by insider attacks. Only about one in five companies (17%) have found a way to better understand insider damage. Forty-four percent of respondents estimate that the cost of an insider breach falls between \$100K and \$2M. Even when breaches cost less than \$100K, the monetary consequences can add up quickly if multiple breaches occur.

► Within your organization, how difficult is it to determine the actual damage of an occurred insider attack?

58%
Moderately
difficult

25%
Very difficult

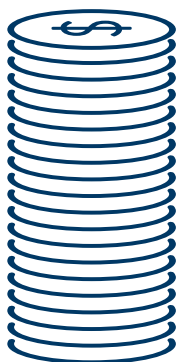


83%

of organizations find it moderately difficult to very difficult to determine the actual damage of an insider attack

► What is the estimated average cost of remediation after an insider attack?

56%



< \$100K

44%

of respondents estimate their attacks cost between \$100K and \$2M

20%



\$100K - \$500K

18%



\$500K - \$1M

6%



\$1M - \$2M

BIGGEST BARRIERS

Cybersecurity teams are drowning in alerts that obscure legitimate risks to data, and they get burned out trying to chase them all down. Therefore, it is no surprise that lack of resources remains the biggest barrier to management of insider threats. Specifically, lack of budget (52%) and lack of staff (45%), followed by lack of suitable tools and technology, and lack of training and expertise (tied at 37%).

► What are the biggest barriers to better insider threat management?



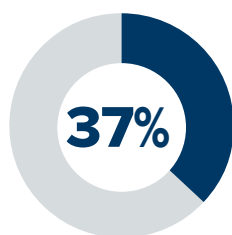
52%

Lack of budget

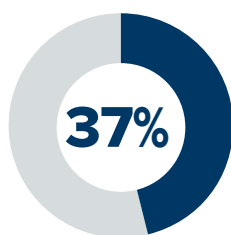


45%

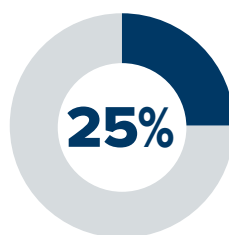
Lack of staff



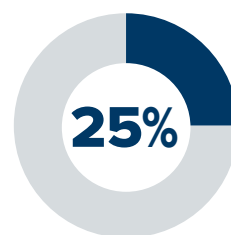
Lack of suitable tools/technology



Lack of training and expertise



Lack of collaboration between separate departments



Not a priority

Disjointed security tools with disparate level of protection 21% | Privacy concerns 19% | Not sure/other 8%

ACCIDENTAL INSIDERS

An important tenet of insider risk is that the majority of data leaks are accidental, not malicious. Regardless of intention, data leaks jeopardize the financial, reputational, or operational well-being of a company, as well as its employees, customers, and partners. Based on feedback from security professionals, there are several ways these accidental breaches occur.

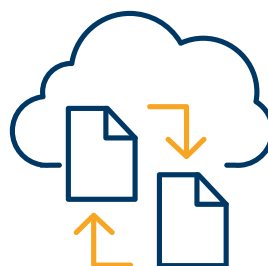
Cybersecurity experts indicate that risky activity like sending business information to personal emails (62%) and copying information to cloud share (45%) are common causes of accidental insider incidents.

► What are the most common accidental insider threats you are most concerned about?



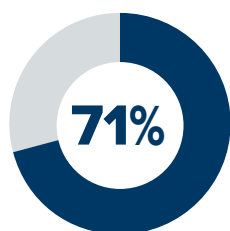
62%

**Sending business
information to
personal emails**

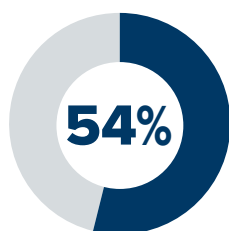


45%

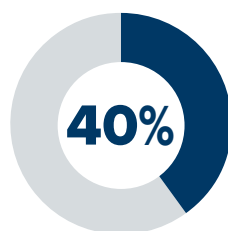
**Copying
information to
cloud share**



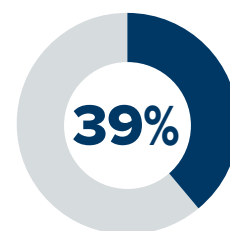
Phishing



Spear-phishing



Poor
passwords



Browsing
suspicious sites

Mobile devices 31% | Misconfigurations 30% | Orphaned accounts 23% | Other 5%

ENABLERS OF INSIDER ATTACKS

The perennial lack of employee training and awareness (37%), increasing number of devices with access to sensitive data (34%), and technology becoming more complex (31%) are the biggest factors contributing to insider threats, including accidental ones.

► What do you believe are the main reasons behind insider attacks?



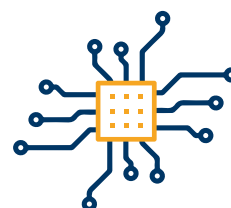
37%

Lack of employee awareness and training



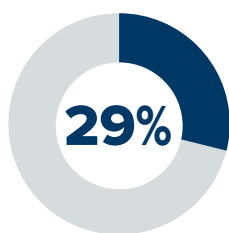
34%

Increasing number of devices with access to sensitive data

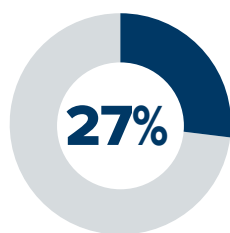


31%

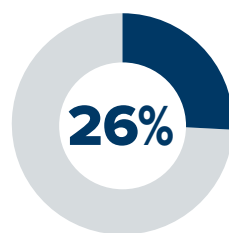
Technology is becoming more complex



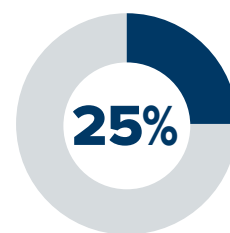
Too many users with excessive access privileges



Increasing amount of sensitive data



More employees, contractors and partners accessing the network



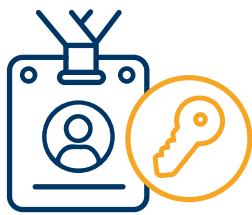
Data increasingly leaving the network perimeter via mobile devices

No consequences for employees (too difficult to prove malicious intent) 23% | Insufficient data protection strategies or solutions 23% | Increasing use of cloud/SaaS apps, cloud infrastructure, and the web 23% | Disgruntled employees/contractors 18% | Attackers are more successful with phishing and social engineering campaigns 15% | Lack of consistent, thorough security policies across SaaS, IaaS, on-prem apps, and the web 14% | Increased public knowledge or visibility of insider threats that were previously undisclosed 12% | Insufficient user monitoring strategies or solutions 10% | Not sure/other 15%

DETECTION AND PREVENTION

Because insiders often have elevated access privileges to sensitive data and applications, it becomes increasingly difficult to detect malicious activity (49%). Combined with the increased use of applications that can leak data (40%) and difficulty to prove malicious intent on part of the employee (30%), the conditions for successful insider attacks are becoming more difficult to control. To protect data from insider threats, security teams need high-fidelity risk detection that cuts through the noise.

► What makes the detection and prevention of insider attacks increasingly difficult?



49%

Insiders already have
credentialed access
to the network
and services



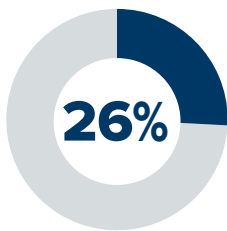
40%

Increased use of
applications that
can leak data
(e.g., Web email,
DropBox, social media)

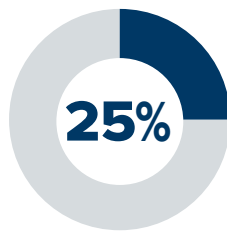


30%

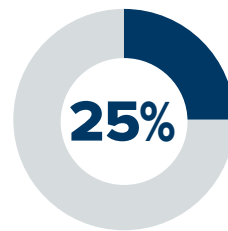
Difficult to prove
malicious intent
on part of the
employee



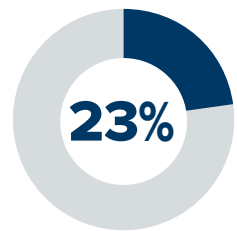
More end-user
devices capable of
theft



Increased amount of
data that leaves
protected boundary/
perimeter



Insiders are more
sophisticated



Personal device
access to corporate
resources

Difficulty in detecting rogue devices introduced into the network or systems 20% | Migration of sensitive data to the cloud along with adoption of cloud apps 20% | Lack of integrated security that protects data consistently across SaaS, IaaS, on-prem apps, the web, and all devices 12% | Challenges related to the management of disjointed and disparate security tools 12% | Absence of an Information Security Governance Program 7% | Lack of relevant cloud security tools 6% | Not sure/other 14%

INSIDER THREAT SOLUTIONS

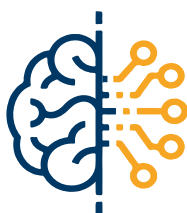
Forty-six percent of organizations have limited capabilities to defend against insider threats. Of the organizations that are defending against insider threats, 27% are using artificial intelligence and machine learning, followed by currently implementing solutions that have been evaluated (24%).

► In what ways are you currently using insider threat capabilities?



46%

We have limited capabilities



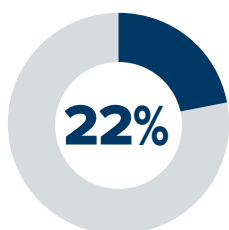
27%

We use artificial intelligence and machine learning

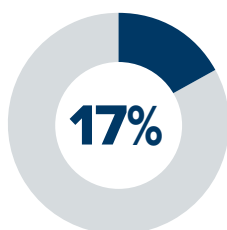


24%

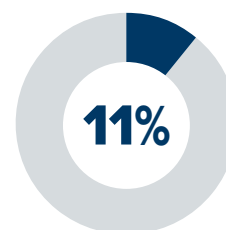
We are currently implementing solutions we have evaluated



We use big data security analytics



We are in the process of evaluating insider threat solutions



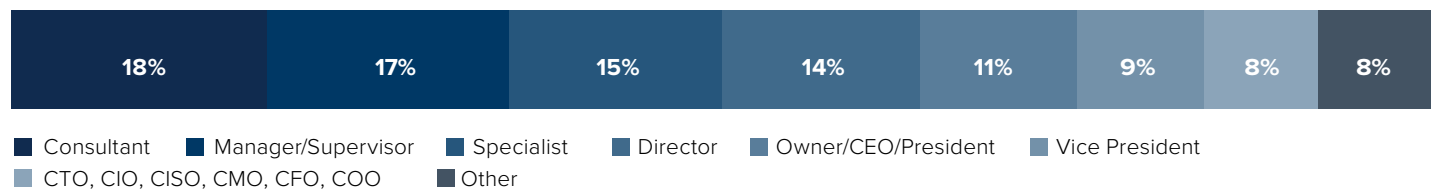
We do not have a solution in place but plan to implement one

The SecOps team identifies threats with a manual process 7% | We do not have a solution in place and have no plans to do so 6%

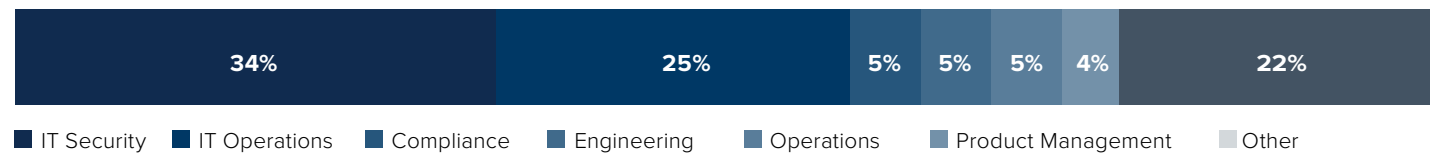
METHODOLOGY & DEMOGRAPHICS

This 2021 Insider Landscape Report is based on the results of a comprehensive online survey of 302 cybersecurity professionals, conducted in June 2021, to gain deep insight into the latest trends, key challenges, and solutions for insider threat management. The respondents range from technical executives to managers and IT security practitioners, representing a balanced cross-section of organizations of varying sizes across multiple industries.

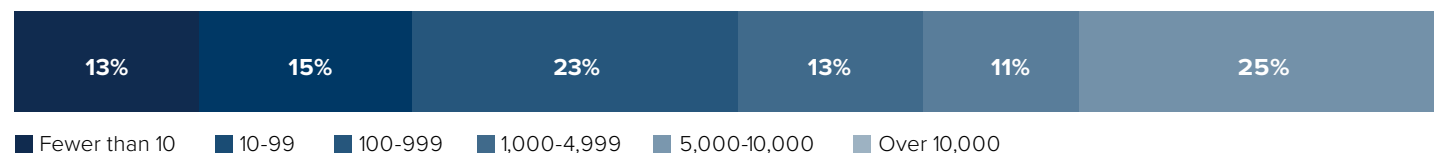
CAREER LEVEL



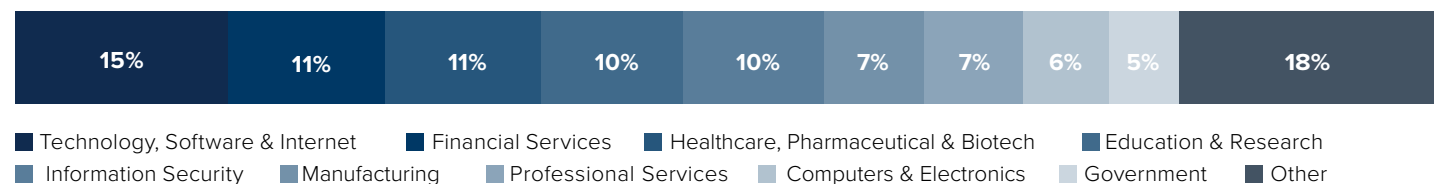
DEPARTMENT



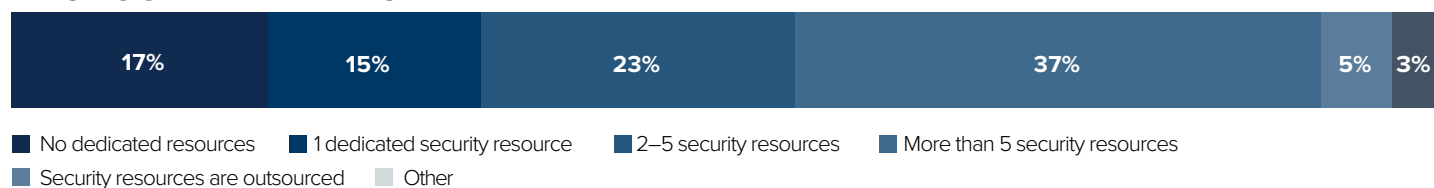
COMPANY SIZE



INDUSTRY



IT SECURITY TEAM SIZE





Code42 is the Insider Risk Management leader. Native to the cloud, the Code42® Incydr™ solution rapidly detects data loss, leak and theft as well as speeds incident response – all without lengthy deployments, complex policy management or blocking employee productivity. With Code42, security professionals can protect corporate data and reduce insider threats while fostering an open and collaborative culture for employees. Backed by security best practices and control requirements, Code42’s insider risk solution is FedRAMP authorized and can be configured for GDPR, HIPAA, PCI and other regulatory frameworks.

More than 50,000 organizations worldwide, including the most recognized brands in business and education, rely on Code42 to safeguard their ideas. Founded in 2001, the company is headquartered in Minneapolis, Minnesota, and backed by Accel Partners, JMI Equity, NewView Capital and Split Rock Partners. Code42 was recognized by Inc. magazine as one of America’s best workplaces in 2020 and 2021.

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